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ATCRBS and Mode S Transponder

Where is the airplane?

CEC300: Computing in Aviation and Aerospace

ATCRBS and Mode S Transponders

- Secondary Radar
  - Instead of using primary radar, which relies on reflected energy to locate a target.
  - Ground Station Interrogates Aircraft (1030 MHz), and Aircraft reply to interrogation (1090 MHz) for mode A, C.
- ATCRBS – Air Traffic Control Radar Beacon system.
  - Mode A – broadcasts IDENT
  - Mode C – broadcasts Altitude
- Mode S – addresses some of the limitations of Mode A and C
  - Additional information transmitted
    - e.g. Squitter and Extended Squitter
  - Supports air-to-air interrogation and replies

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ATCRBS – Air Traffic Control Radar Beacon System

Developed in 1956, ATCRBS was the first air route surveillance radar system developed and purchased for the purposes of air traffic control. The technology was based closely on that of the military's IFF (Identification Friend or Foe) system.

Interrogation from ground– 1030 MHz
Reply from airplane – 1090 MHz

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ATCRBS – Air Traffic Control Radar Beacon System

On the ground:

In the airplane:

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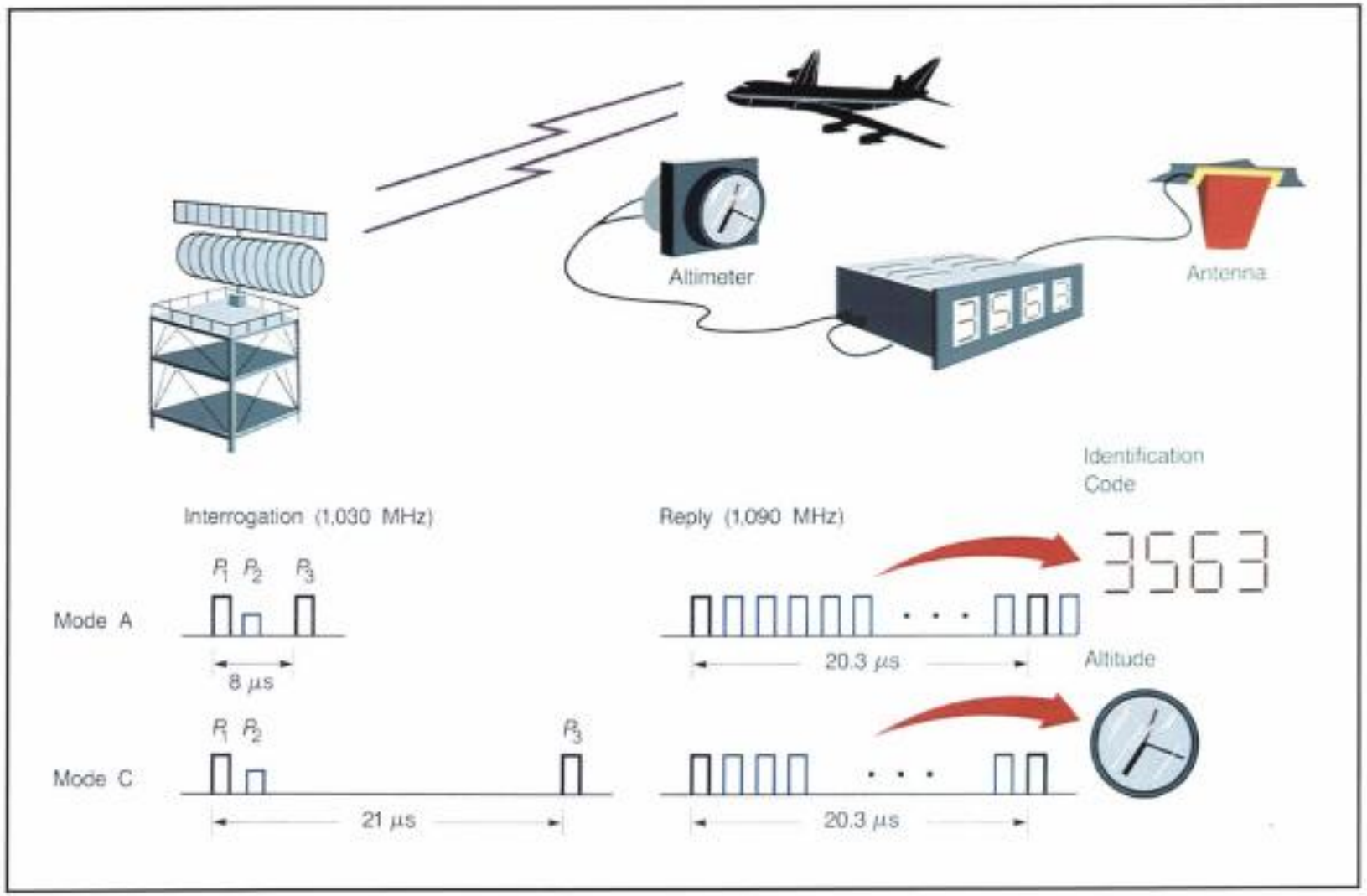

Fig. 1—Air Traffic Control Radar Beacon System (ATCRBS).

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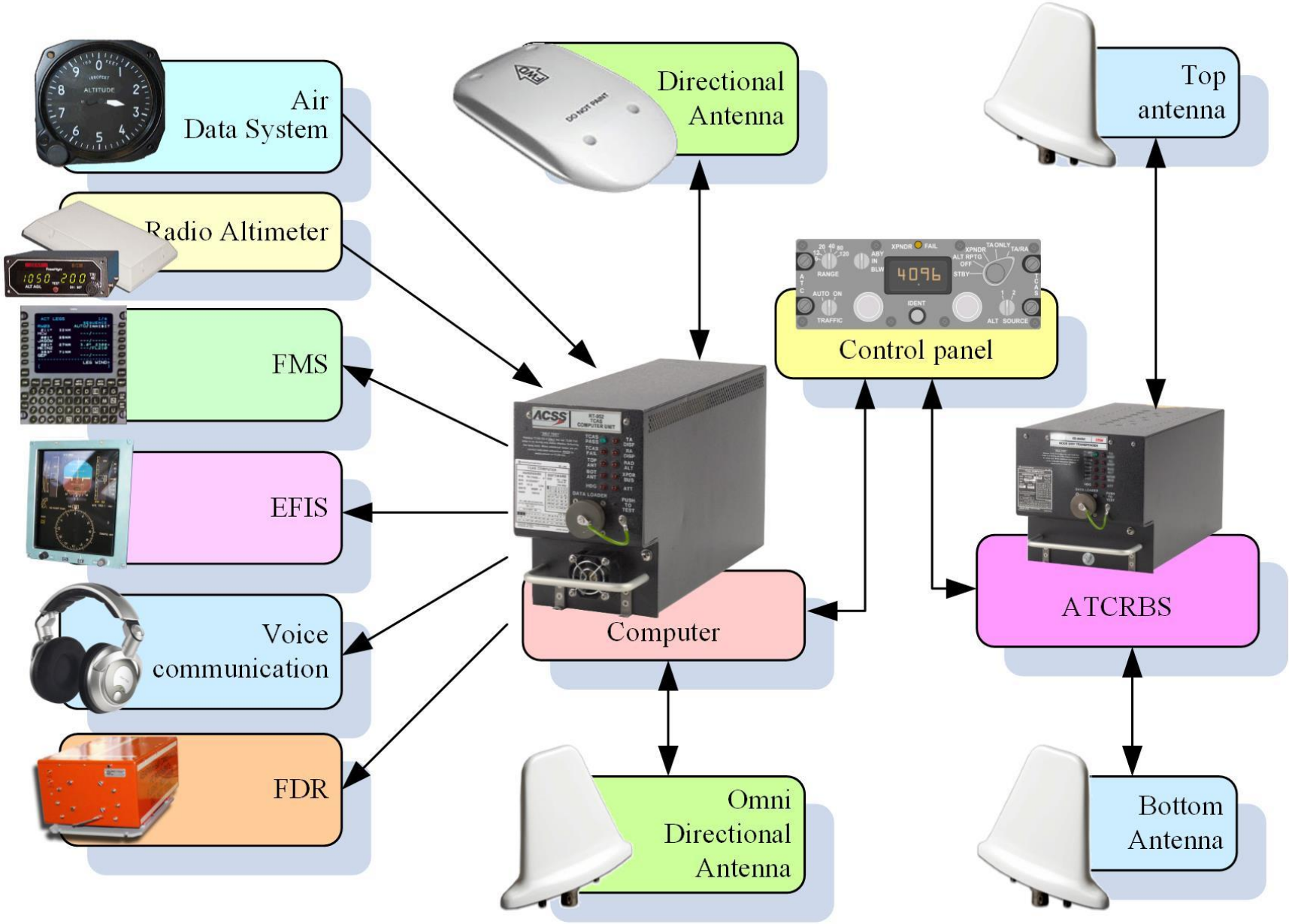

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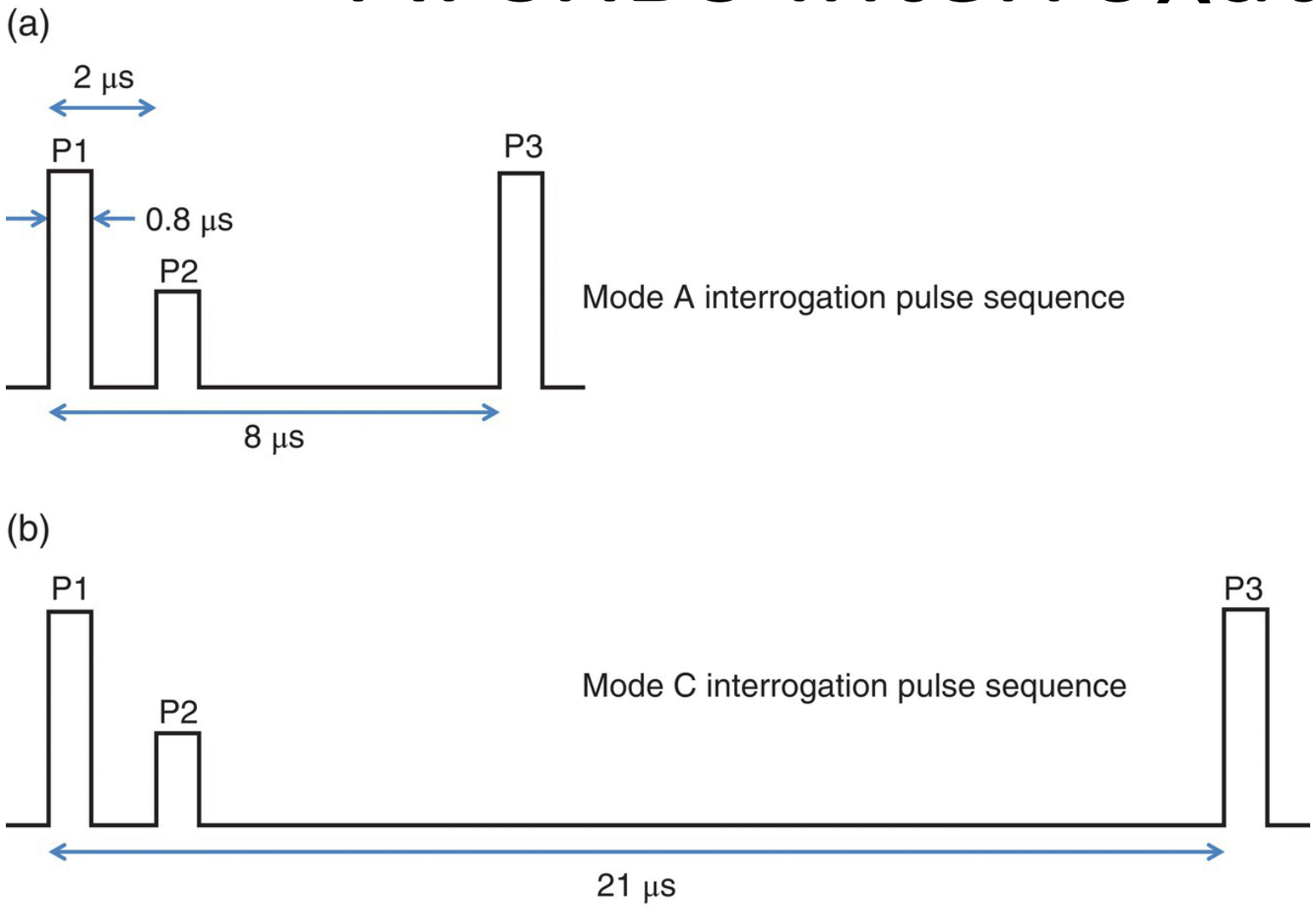
ATCRBS Interrogations



(from Binns, 5 Primary and Secondary Radar | Aircraft Systems)

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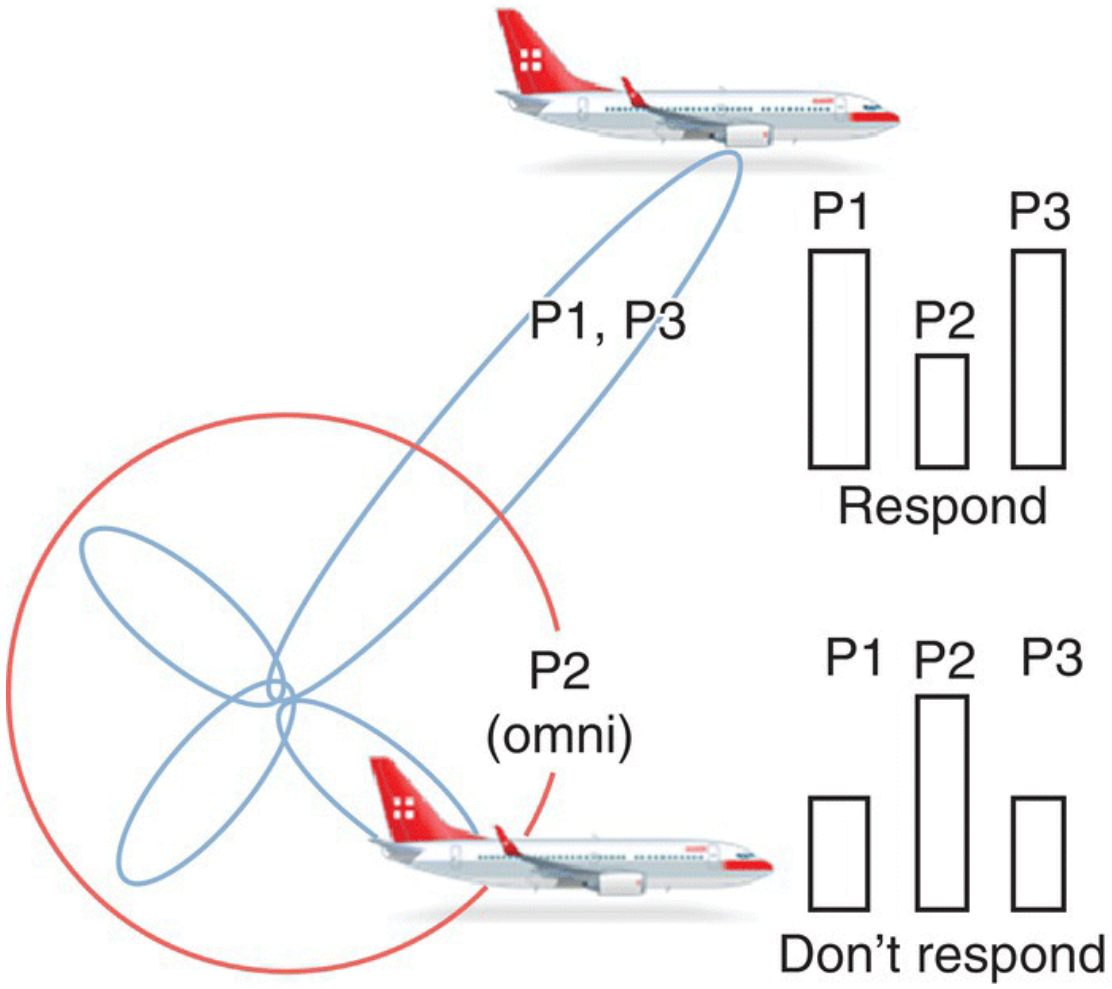
Sidelobe Suppression Pulse



Omni directional antenna for P2 has a power greater than the energy of the sidelobes.

If P1 > P2, then Respond

Otherwise, ignore

(from Binns, [5 Primary and Secondary Radar | Aircraft Systems])

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When Reply to Interrogation

ATCRBS Transponder replies to an interrogation so long as:

1. Transmission of P1 to P2 is $2\mu s$ apart
2. P1 > P2 signal
3. Delay of 8 μs if Mode A or $21\mu s$ if Mode C from P1 rising edge to rising edge of P3

Once a valid interrogation is received, the turnaround time for reply is $3\mu s$.

The next slide reviews the reply message and its respective timing.

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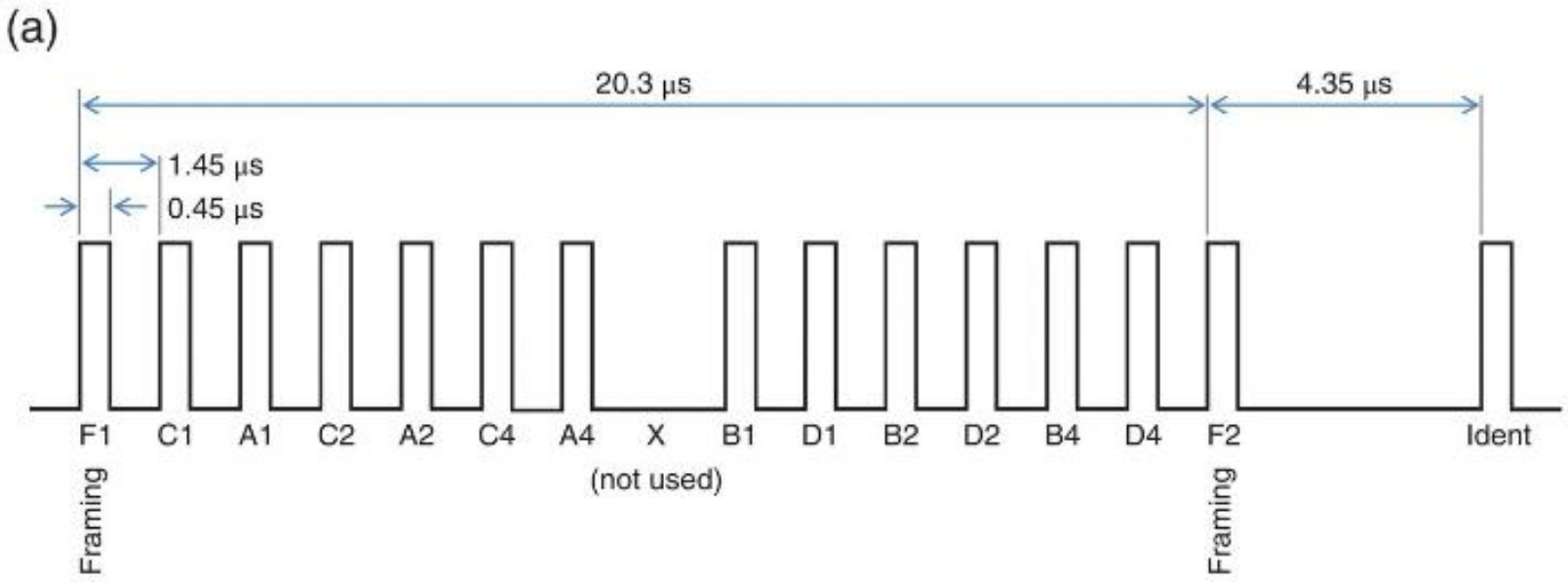
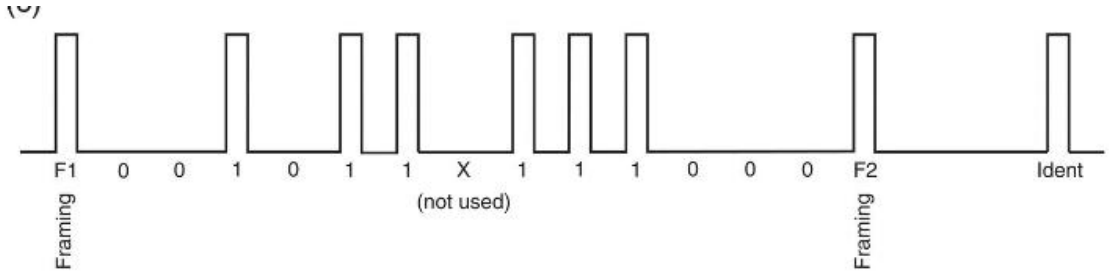
(from Binns, 5 Primary and Secondary Radar | Aircraft Systems)

ATCRBS Reply Format



Transmitted code 4361

4	2	1	
1	0	0	A
0	1	1	B
1	1	0	C
0	0	1	D

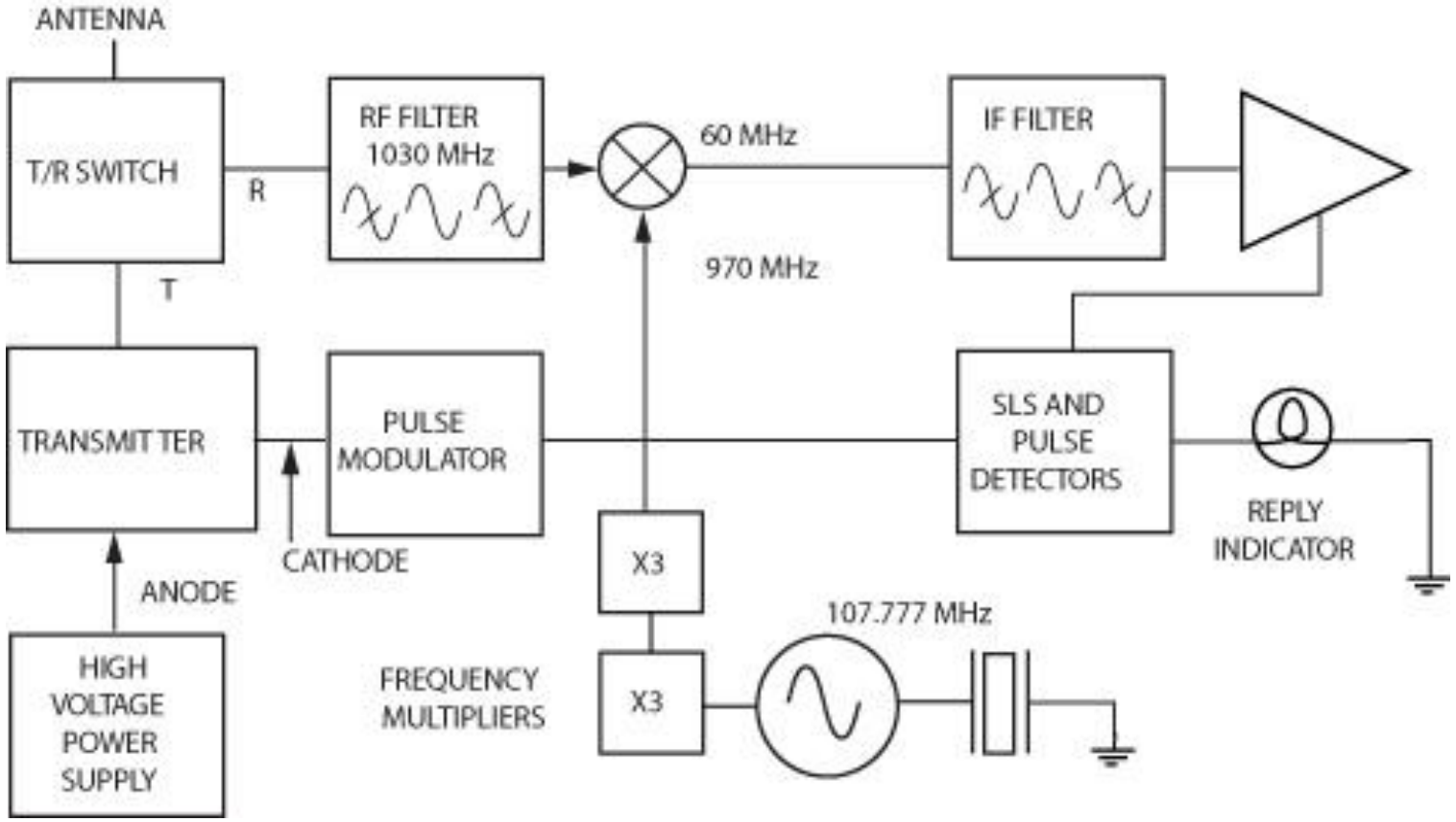


ATCRBS Transponder Mount Types

File:Transponder 2.jpg - Wikimedia Commons

NXT-600/-700/800™ | Mode S Transponders | L3Harris

ATCRBS Transponder Block Diagram



Source: Helfrick, 2010

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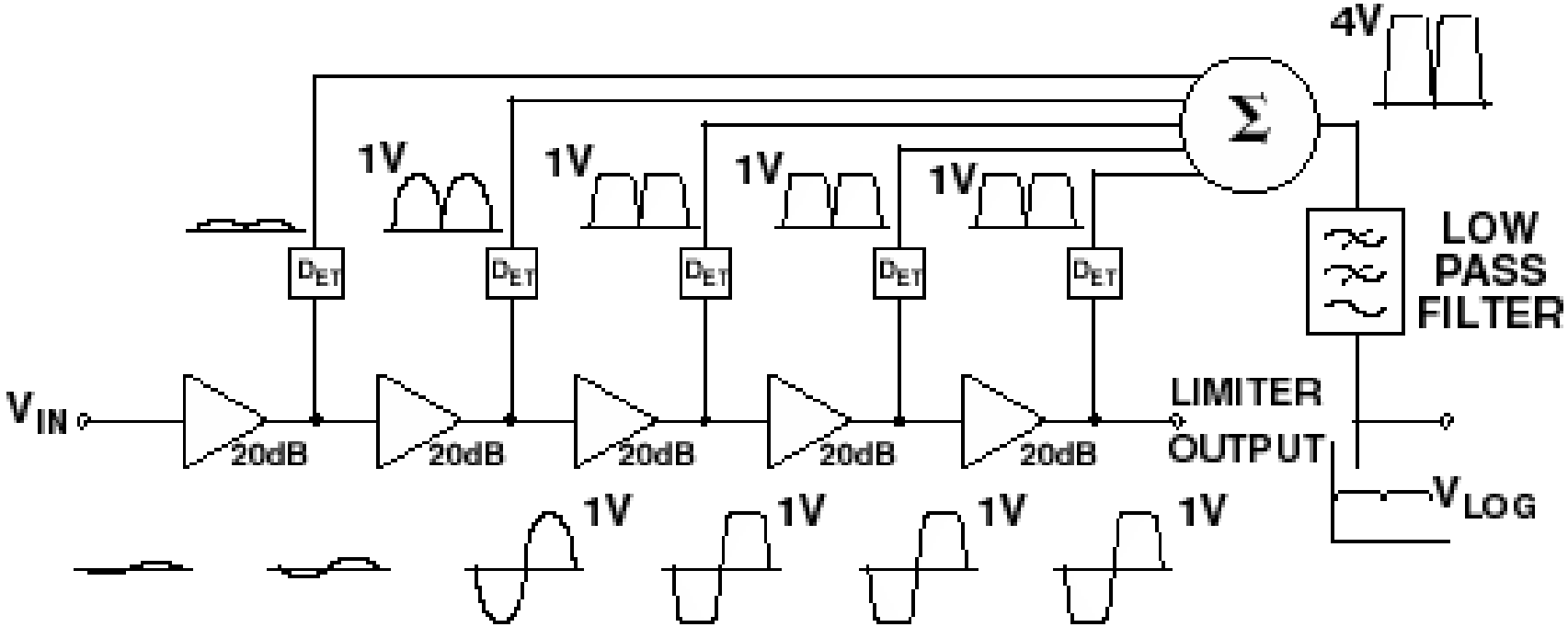
Logarithmic Amplifier



***Purpose**: Obtain similar output for extreme changes in input power*

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Transponder Sensitivity and Power Output

Transponder Minimum Triggering Level is Set to -73 dBm +/- 4 dB at the Transponder Input/Output Connector

Power Output for Class 1A or 2A ATCRBS Transponders is 125 Watts

Power Output for Class 1B and 2B ATCRBS Transponders is 70 Watts

No Class of Transponder may Exceed 500 Watts

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Encoding Altimeter

- An encoding altimeter is used to transmit the barometric altitude of an aircraft for a Mode C and Mode S transponders.
- Mode C – altitudes are encoded in 100 ft increments.
- Mode S – altitudes are encoded in 25 ft increments
- Gilham Codes – a type of “gray code” is used to encode the altitude into the appropriate 11-bit sequence

The Gillham Code

It is used to transmit uncorrected barometric altitude between an encoding altimeter or analog air data computer and a digital transponder.

A form of Gray code
Range: -1200’ to 126,700’
100’ increments
Uses 11 of the available 12 bits
D1 is not used

Altimeter Encoder

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The Gillham Code

Gillham binary code [D124 A124 B124 C124]	Squawk octal code [ABCD]	Height [m]	Height [ft]
000 000 000 001	0040	−365.76	−1200
000 000 000 011	0060	−335.28	−1100
000 000 000 010	0020	−304.8	−1000
000 000 000 110	0030	−274.32	−900
000 000 000 100	0010	−243.84	−800
000 000 001 100	0410	−213.36	−700
000 000 001 110	0430	−182.88	−600
000 000 001 010	0420	−152.4	−500
000 000 001 011	0460	−121.92	−400
000 000 001 001	0440	−91.44	−300

…

010 000 000 110	0032		126400
010 000 000 010	0022		126500
010 000 000 011	0062		126600
010 000 000 001	0042		126700

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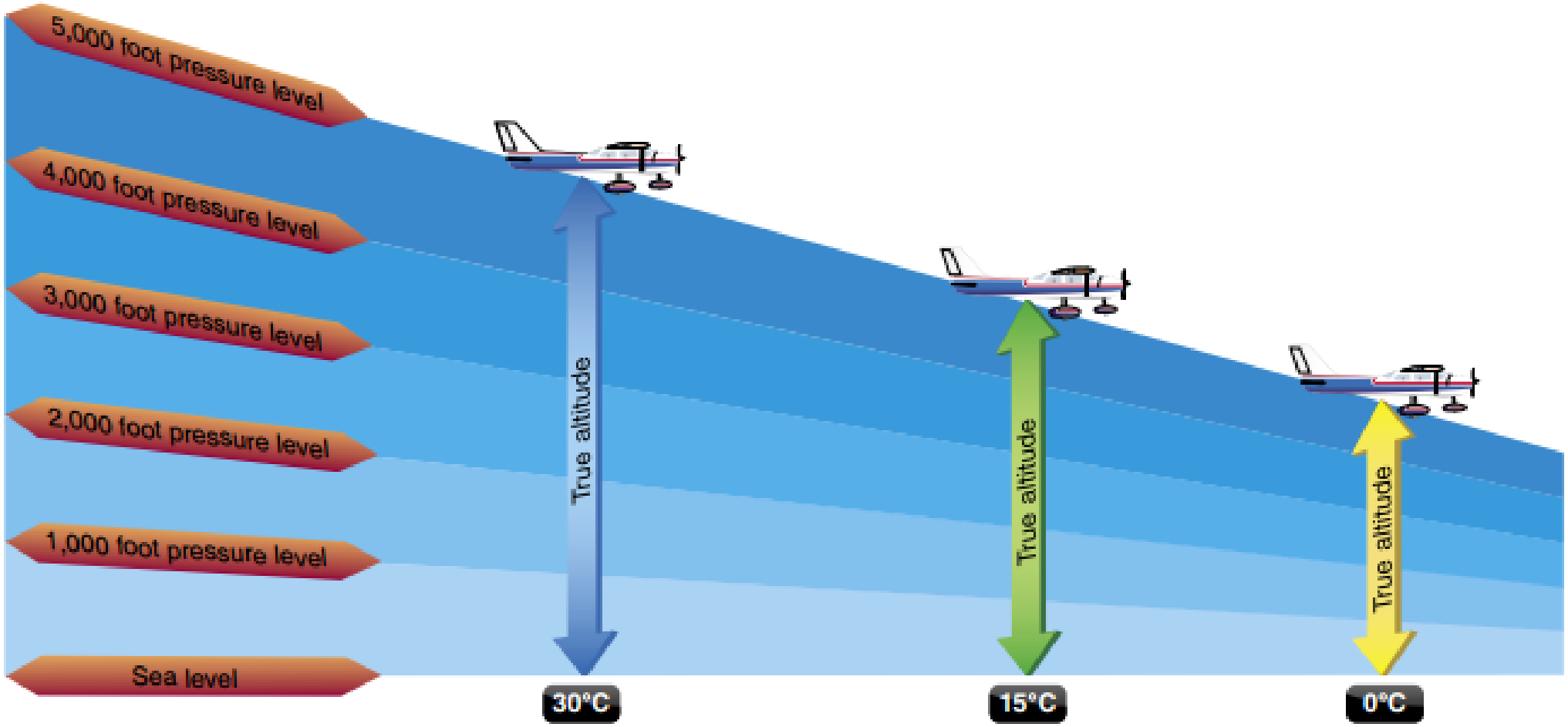
The Need to Use Pressure Altitude



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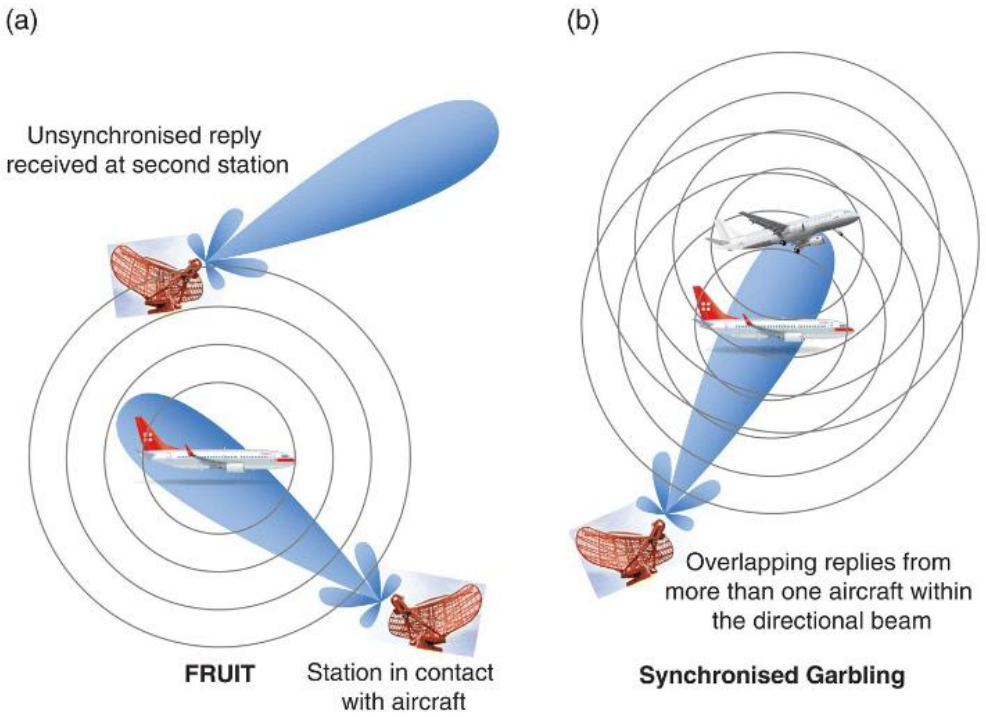
Issues with ATCRBS

- Garble / Fruit
  - Wasteful of Spectrum
  - Required Repeated Interrogations
- Limited Info: altitude and IDENT
- No guarantees of receipt or integrity (no means to mitigate TX error)



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Mode S Technology attempts to mitigate ATCRBS

- All Call to request all aircraft to reply with a specified probability that they should respond, e.g. ½, ¼, 1/8, 1/16 chance to reply
  - After several attempts, the majority of traffic should be detected at the ground station.
  - Stochastic acquisition

- Selective Interrogations (aka “roll call” – once local aircraft are known, interrogate those aircraft.

- Lock out aircraft that have responded with updates to avoid unnecessary replies to all-call.

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Mode S (Select)

- Developed to provide more modern enhancements and reduce the amount of over interrogation occurring.
- Built to be backward compatible with ATCRBS air/ground systems.
- Common Features with ATCRBS:
  - 1030 MHz (interrogation) and 1090 MHz (reply)
  - Vertically polarized
  - Same signal levels

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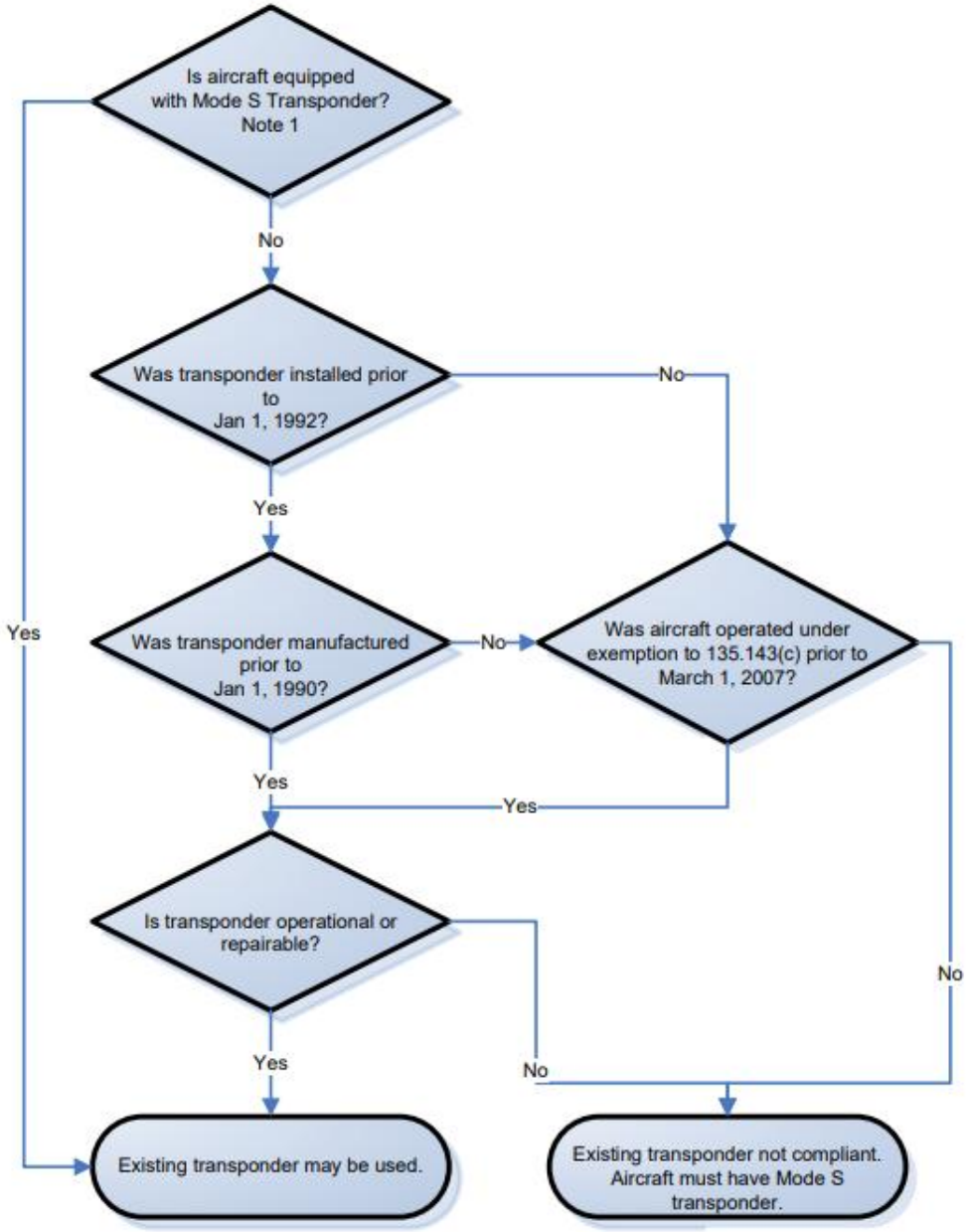
Mode Select, Mode-S, Transponder



Note 1: A transponder is not required for operation under Part 135 however the operator must comply with the requirements of 14 CFR 91.215(b)

ICAO 24-Bit Aircraft Address

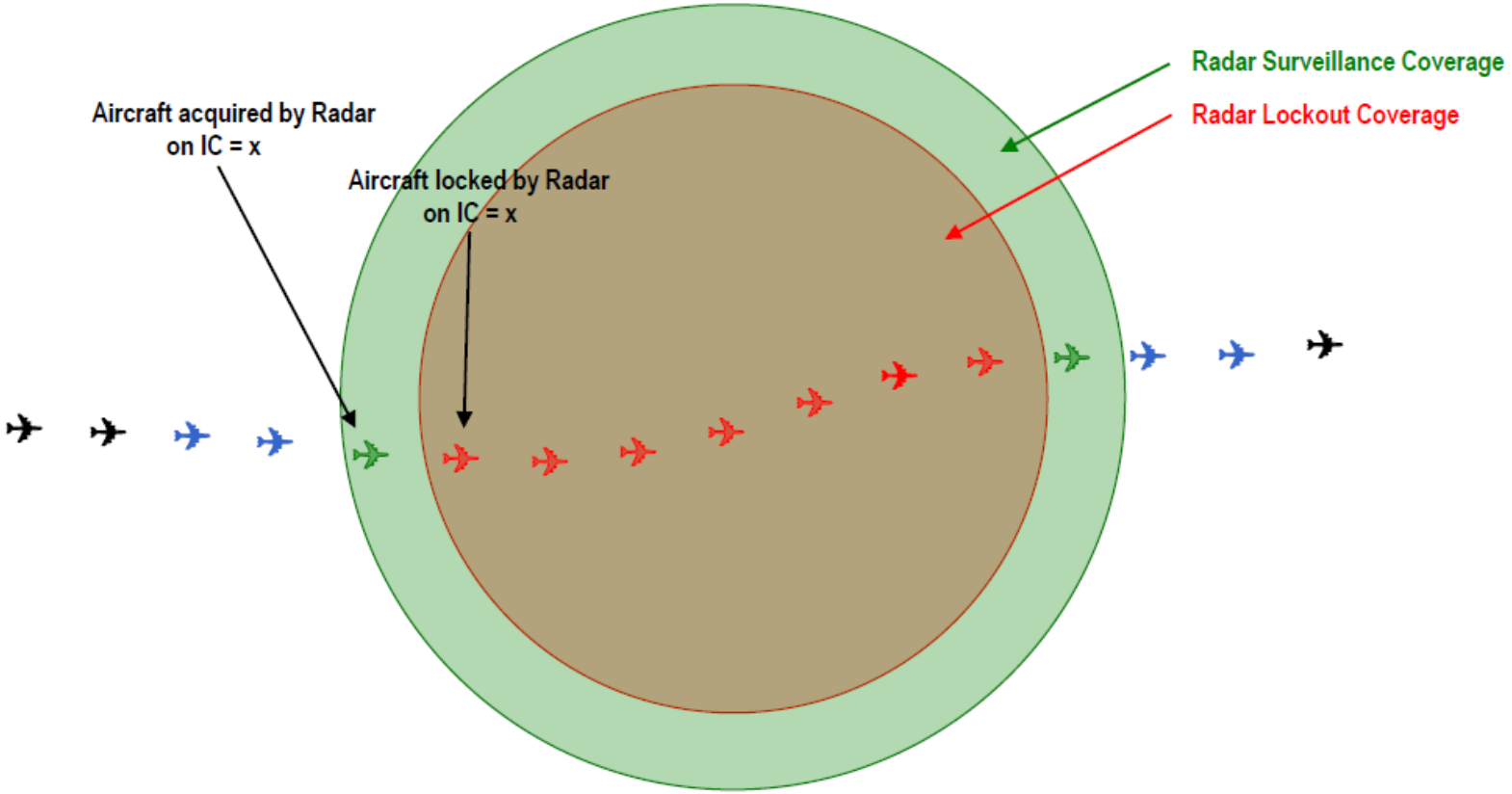
- Instead of using ATC assigned transponder codes…
- Registered aircraft are assigned a 24-Bit address (ICAO Address)
- With 24-bit, enough addresses to support 2^24 or approx. 16.7 million

https://www.icao.int/WACAF/Documents/Meetings/2023/Workshop-24-Bits/Guidelines%20for%20ICAO%2024%20Bits%20Adresses%20Assignment.pdf

Annex 10 — Aeronautical Communications Volume III

State	Number of addresses in block					Allocation of blocks of addresses (a dash represents a bit value equal to 0 or 1)					
	1 024	4 096	32 768	262 144	1 048 576						
Democratic People's Republic of Korea			*			0111	00	100	---	--	----------
Democratic Republic of the Congo		*				0000	10	001	100	--	----------
Denmark			*			0100	01	011	---	--	----------
Djibouti	*					0000	10	011	000	00	----------
Dominican Republic		*				0000	11	000	100	--	----------
Ecuador		*				1110	10	000	100	--	----------
Egypt			*			0000	00	010	---	--	----------
El Salvador		*				0000	10	110	010	--	----------
Equatorial Guinea		*				0000	01	000	010	--	----------
Eritrea	*					0010	00	000	010	00	----------
Estonia	*					0101	00	010	001	00	----------
Ethiopia		*				0000	01	000	000	--	----------
Fiji		*				1100	10	001	000	--	----------
Finland			*			0100	01	100	---	--	----------
France				*		0011	10	---	---	--	----------

Aircraft acquisition, selective interrogations and lock-out



Aircraft not in line of sight of radar or not in power budget → does not receive All-Call interrogations (broadcast)

Aircraft outside surveillance coverage → receives All-Call interrogations (broadcast) and replies, but replies not processed by radar

Aircraft acquired by radar in surveillance coverage → selective interrogations (Roll-Call)
→ not locked: receive All-Call interrogations and replies

Aircraft locked by radar in lockout coverage → does not reply to All-Call interrogations

Image Source: Bodart, 2019

All-Call and Roll Call

All-Call – identifies aircraft within the vicinity (1/3 of interrogations)
Roll-Call - Interrogation identified aircraft for data (2/3 of interrogations)

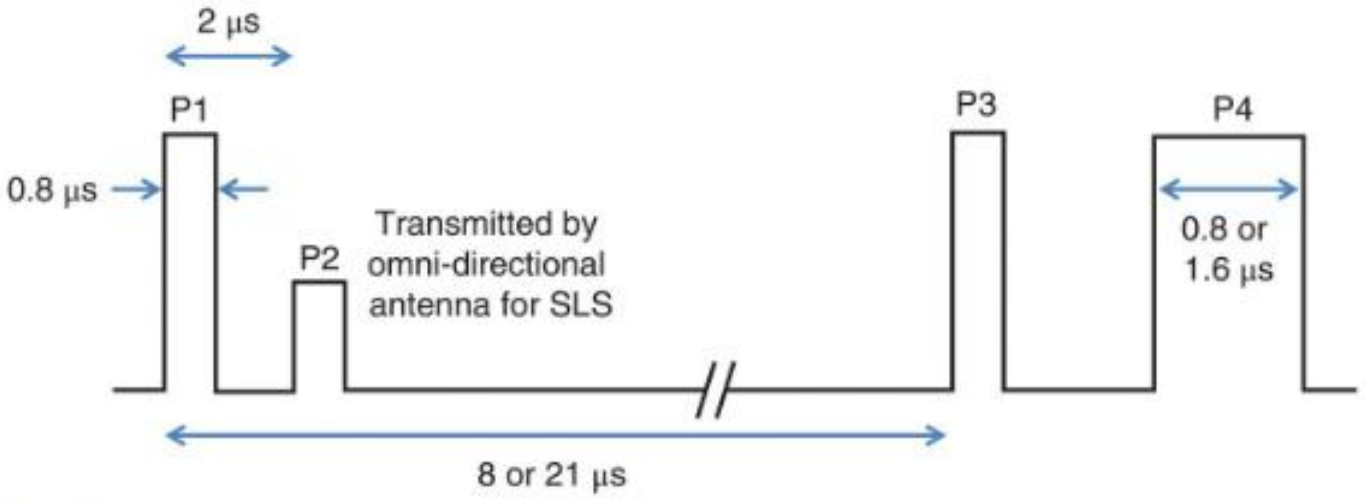


Figure 5.21 Format of the mode S all call signal from the ground station.

(Image source: 5 Primary and Secondary Radar | Aircraft Systems)

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(Image source: 5 Primary and Secondary Radar | Aircraft Systems)

All-Call Signal Timing

- Signal transmitted with P1, P2, and P3 (like ATCRBS)
- Additional P4 signal alternates between 0.8 μs to 1.6 μs pulse width (all other pulses in the interrogation are 0.8 μs
  - 0.8 μs ignored by Mode S, but will trigger Mode A and C transponders
  - 1.6 μs triggers Mode S transponders to respond
  - Enables backward compatibility
- Reply includes ICAO address and interrogators identifier.
- Aircraft replies that are received without garble are considered “locked” or “acquired”.

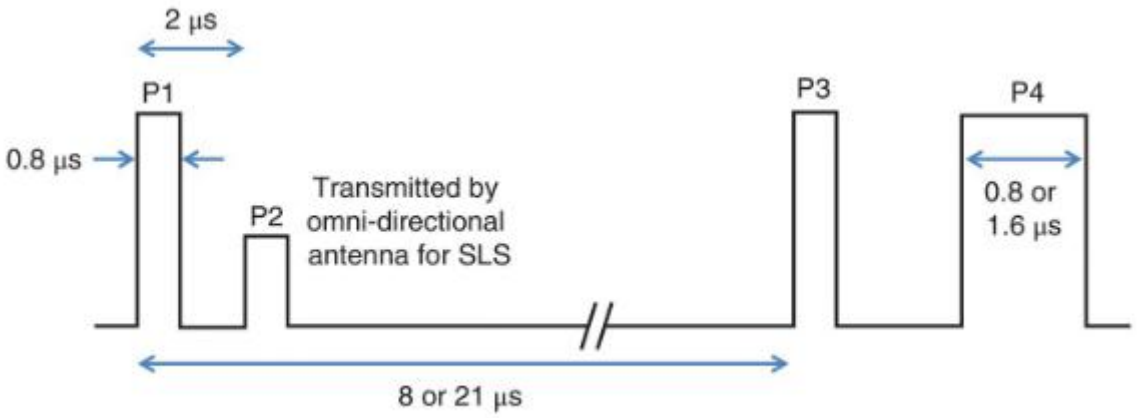

Figure 5.21 Format of the mode S all call signal from the ground station.

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Selective Call Interrogation "Roll Call"

- Pulse format
- When P1 and P2 are the same strength (Mode A/C ignore)
- Reference pulse of 1.25 μs issued at start of data word called "sync"
- Data word is either 56 bits or 112 bits (between 0.5 μs framing pulses)
- 0.25 μs with per bit
- Data block includes
- 24-bit ICAO address
- Interrogator identity
- All call if ICAO address all 1s

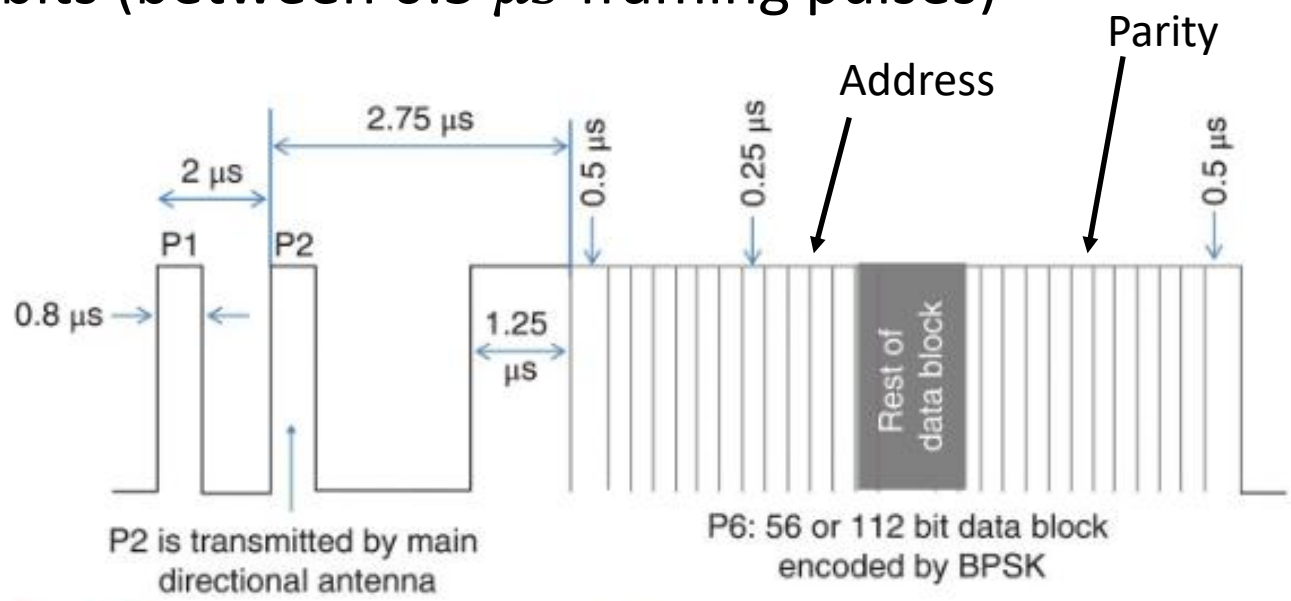
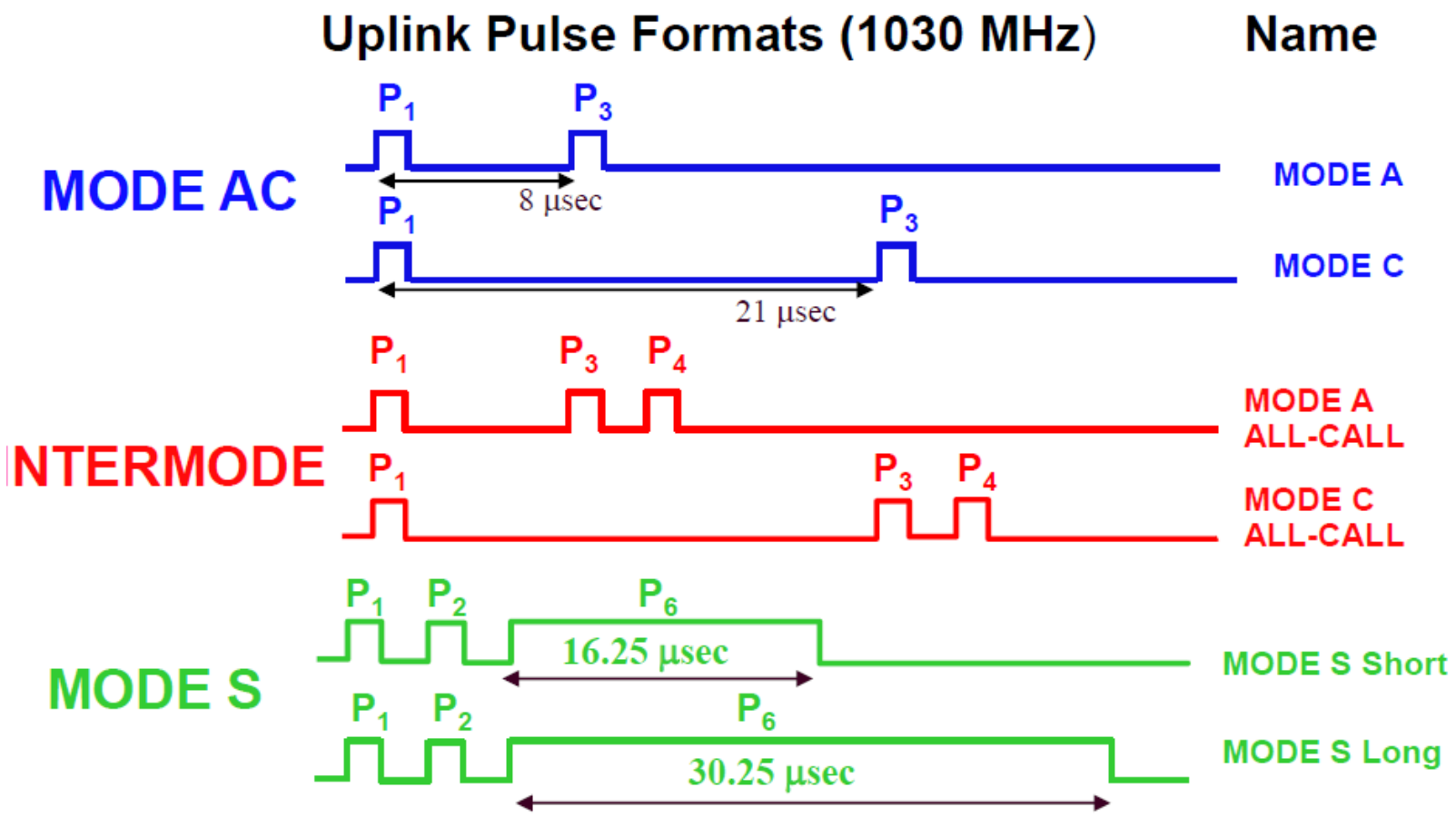


Figure 5.22 Format of the mode S selective call signal to a specific aircraft from the ground station. The P2 pulse is transmitted by the main directional antenna and has the same strength as the P1 pulse so the interrogation is ignored by mode A and mode C aircraft.

(Image Source: [5 Primary and Secondary Radar | Aircraft Systems])

Mode S and Mode AC interoperability



Mode-AC transponder	Mode-S transponder
MODE A	MODE A
MODE C	MODE C
MODE A	No Reply
MODE C	
No Reply	MODE S if addressed or All-Call

Transponder Replies (1090 MHz)

Image Source: Bodart, 2019

Locking Out

- After All-Call, Mode S radar will only interrogate the known aircraft with the SELECT(Roll Call) interrogation.
- Lockout message sent with the first SELECT Message.
- Aircraft transponders lock out replies to the station's ALL Call interrogations until SELECT Message indicating lock out has ended.

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Stochastic Acquisition

- Mitigate garbling by not having each aircraft respond to every all call.
- With less aircraft responding, there is less likelihood of garble
- Set by the ground-station's interrogation, the aircraft are told to respond with one of the following probabilities: 100%, 50%, 25%, 6.25%, 0%.
- After several all-calls the neighboring aircraft should be acquired.

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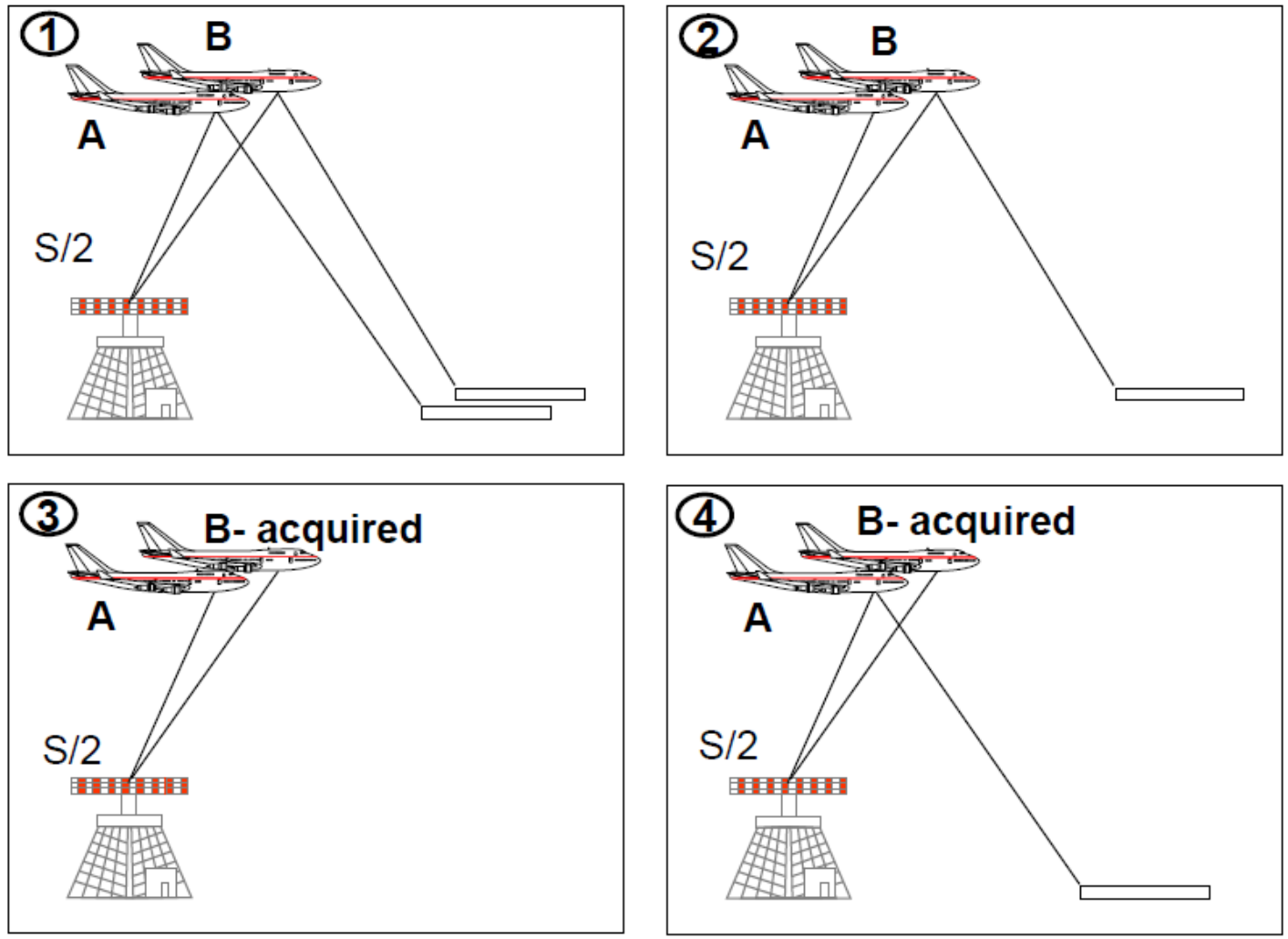
Stochastic Acquisition

Image Source: Bodart, 2019



Stochastic S/2 Interrogations with 50% set as PR

1. A and B both reply
   Replies overlap in time
   Both are garbled and lost

2. A decides No Reply (50%)
   B replies
   B acquired and locked out

3. B is locked out
   A decides No Reply (50%)

4. B is locked out
   A replies
   A acquired and locked out

Mode S, Select Reply

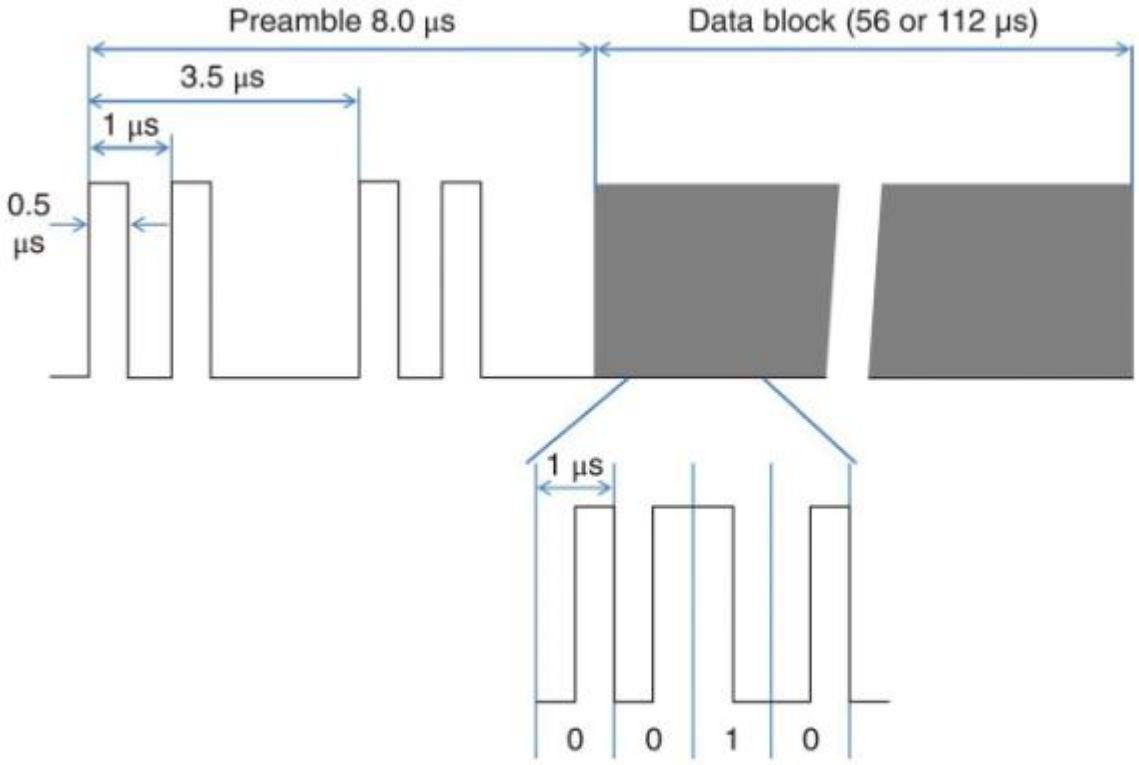


Figure 5.23 Format of the mode S reply.

Data block uses pulse position modulation.

Reply is 56 or 112 microseconds long

Data block contains

- All Messages - aircraft address, parity, control bits
- Long Message – Additional Flight Information

Weather

(Image Source: 5 Primary and Secondary Radar | Aircraft Systems)

Squitter

Aircraft replying to nothing.
It broadcasts its:

- Same preamble as Mode S reply
- 8-Bit control code,
- 24-Bit aircraft address
- 24-bit parity

Extended Squitter will be discussed with ADS-B

(Image Source: 5 Primary and Secondary Radar | Aircraft Systems)

Reducing Errors

- **Parity Check:**
  - Cyclic Redundant Check (CRC) used to provide a sequency of parity bits, which are appended to the transmitted message.
    - Polynomial, iterative XOR process
  - Receiver can perform same CRC then compare the results.

- In response, the system can re-interrogate the aircraft that produced an error message.

- ATC can be notified of aircraft with recurring errors (might be faulty or dealing with local interference impacting ground or aircraft).

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Air-to-Air Mode S

An advantage of Mode S is that it can support not only ground-to-air SSR.

It also supports Air-to-Air!

- Aircraft can send ALL CALL and suitably equipped aircraft will respond (Mode A, Mode C, and other Mode S).
- With aircraft acquired, it can then use the SELECT message to request updates from the detected aircraft.
- Having antenna on top and bottom of aircraft enable trans and receipt to aircraft above it.

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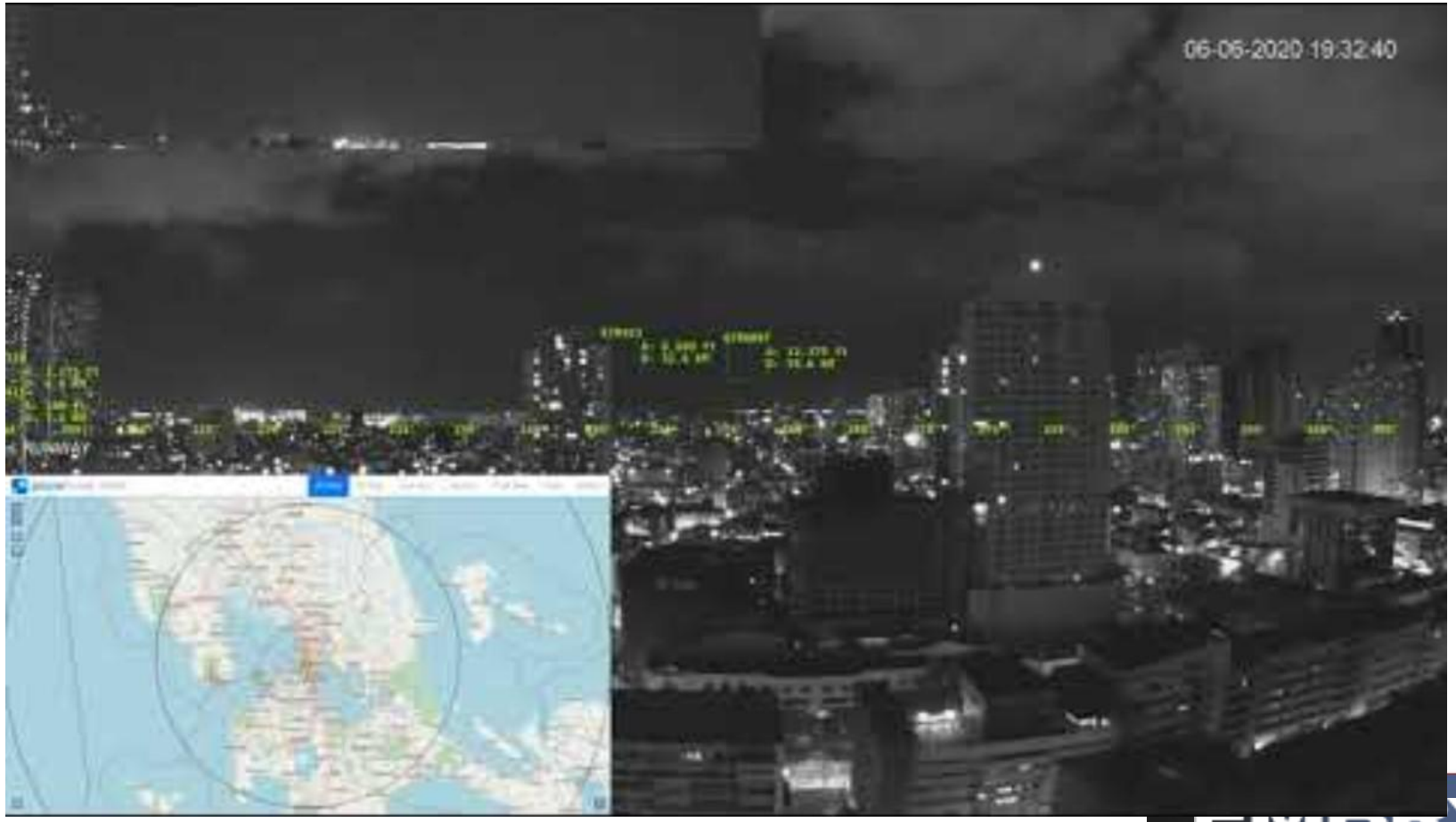
Example of Mode S Hardware

Mode S – Radar Display



Mode-S [ADS-B] (SSR) Radar Camera Overlay

Antenna Configuration

- Blade antennas are installed on the top and bottom of the aircraft
- Each has its own receiver and decoder.
- Best signal wins – the signal with the least error is selected (via error checking). Tie break: Pick the signal with the highest signal strength.

Figure 5.24 Pair of blade antennas on the belly of a light twin used to provide mode S traffic surveillance.

(Image Source: 5 Primary and Secondary Radar | Aircraft Systems)

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Review

1. Describe the function of a transponder in an SSR system.
2. Explain the purpose(s) of the P2 pulse in Mode A and Mode C interrogations, and how are they different for Mode S?.
3. How does the aircraft determine if an interrogation pulse sequence is coming from a side lobe?
4. What information does a Mode A reply from an aircraft provide?
5. What additional information is included in a Mode C reply, compared to a Mode A reply?
6. How did Mode S aim to improve upon the limitations of Mode A and Mode C systems?
7. Describe the three fundamental interrogation modes of Mode S.
8. Explain how Mode S enables traffic surveillance independent of primary radar.

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